Additional Problems: Newsvendor

1. The manager for Taylor Swift needs to choose a venue for her upcoming North America tour. An analysis of the past ticket sales show that the distribution of demand can be modeled as a continuous uniform distribution between 50,000 and 70,000 attendees. The average selling price of a seat is $240 whereas the cost of a seat is $50 per seat. How many seats should a venue for the next concert have?

Solution. We have the following cost parameters: $\text{Unit cost} = \$50/\text{unit}$; $\text{selling price} = \$240/\text{unit}$; and $\text{salvage revenue} = \$0/\text{unit}$

Consequently, the unit overage cost is given by

$$c_o = 50 - 0 = \$50/\text{unit}$$

Similarly, the unit underage cost is given by

$$c_u = 240 - 50 = \$190/\text{unit}$$

Then, the critical ratio is given b

$$F(q^*) = \frac{c_u}{c_u + c_o} = \frac{190}{190 + 50} = 0.7917 \Longrightarrow F(q^*) = P\{X \leq q^*\} = 0.7917$$

Recall that the pdf for a continuous uniform distribution $X \sim U(a, b)$ is given by

$$f(x) = \begin{cases} \dfrac{1}{b-a} & a \leq x \leq b \\ 0 & otherwise \end{cases}$$

and its cdf is given by

$$F(x) = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \leq x \leq b \\ 1 & b < x \end{cases}$$

Hence, using interpolation, we can determine the optimal number of seats, q^*, as follows:

$$\frac{q^* - 50{,}000}{0.7917 - 0} = \frac{70{,}000 - 50{,}000}{1 - 0} \Rightarrow q^* = 50{,}000 + (0.7917)\frac{20{,}000}{1} = 65{,}834 \text{ units}$$

Consequently, we have $q^* = 65{,}384$ seats.

2. The number of the magazine Interior Design sold at the University of Florida bookstore follows a Poisson distribution with an expected number of 4.4 units per month. The purchase cost for the magazine is $15.00 and the sale price is 30.00. Suppose that, by the end of the month, the bookstore can sell all the magazines left over to a second-hand bookstore at a discounted price

of 8.00 per unit. What's the optimal number of magazines that the bookstore should order per month?

Solution. We have the following cost parameters: Unit cost $= \$1.59/\text{unit}$; $p = \$2.19/\text{unit}$; and $v = \$0.99/\text{unit}$

Consequently, the unit overage cost is given by

$$c_o = 15 - 8 = \$7/\text{unit}$$

Similarly, the unit underage cost is given by

$$c_u = 30 - 15 = \$15/\text{unit}$$

Then, the critical ratio is given b

$$F(q^*) = \frac{c_u}{c_u + c_o} = \frac{15}{7 + 15} = 0.6818 \Longrightarrow F(q^*) = P\{X \leq q^*\} = 0.6818$$

Recall that the probability mass function (pmf) of a random variable that is Poisson distribution Pois $X \sim Pois(\lambda)$ is given by

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Since the demand is Poisson distributed, we need to tabulate the number of magazines sold, denoted by X, and the associated probability using the pmf given above until the cdf value exceeds the critical ratio value of 0.6818.

$X = k$	$P(X = k)$	$P\{X \leq k\}$	Stop?
0	$\frac{4.4^0 e^{-4.4}}{0!} = 0.0123$	0.0123	No
1	$\frac{4.4^1 e^{-4.4}}{1!} = 0.0540$	0.0663	No
2	$\frac{4.4^2 e^{-4.4}}{2!} = 0.1188$	0.1851	No
3	$\frac{4.4^3 e^{-4.4}}{3!} = 0.1743$	0.3594	No
4	$\frac{4.4^4 e^{-4.4}}{4!} = 0.1917$	0.5512	No
5	$\frac{4.4^5 e^{-4.4}}{5!} = 0.1687$	0.7199	Yes

Consequently, the optimal decision is to order $q^* = 5$ magazines each month.

3. Northwest Seafood in Gainesville buys fresh mussels daily for $4.20 per two-pound bag and sells them for $5.70 per bag. At the end of each business day, any mussels remaining are sold to a cat food producer for $2.40 per bag. If the daily demand for mussels can be approximated by a normal distribution with a mean of 40 bags and a standard deviation of 5 bags, what is the optimal order quantity (in two-pound bags) for mussels?

Solution: We have the following cost parameters: c = $4.20/unit, p = $5.70/unit, and v = $2.40/unit. Consequently, the unit overage cost is given by

$$c_o = 4.20 - 2.40 \ = \$1.80/\text{unit}$$

Similarly, the unit underage cost is given by

$$c_u = 5.70 - 4.20 = \$1.50/\text{unit}$$

Then, the critical ratio is given by

$$F(q^*) = \frac{1.50}{1.50 + 1.80} = \ 0.4545.$$

We have $F(q^*) = P\{X \leq q^*\} = 0.4545$. Note that $X \sim N(40, 5^2)$. Consequently, we have

$$F(q^*) = \ 0.4545 \ \Rightarrow \ \Phi(z) = \ 0.4545 \ \Rightarrow \ z^* \ = \ -0.1143$$
$$q^* \ = \ \mu \ + \ z^*\sigma$$
$$= \ 40 \ + \ (-0.1143)(5)$$
$$= \ 39.43 \text{ bags}$$

4. Bonnie's is a small grocery store that sells fresh produce that is acquired from a local farmer. During the strawberry season, the demand for fresh strawberries can be reasonably approximated using a normal distribution with a mean of 40 quarts per day and a standard deviation of 6 quarts per day. Bonnie estimates the unit cost of overage to be 35 cents per quart.

a. If Bonnie orders 49 quarts per day, what is the implied unit cost of underage per quart?
b. If the daily demand for mussels can be approximated by a normal distribution with a mean of 40 bags and a standard deviation of 5 bags, what is the optimal order quantity (in two-pound bags) for mussels?

Solution: We have $q^* \ = \ 49$ boxes. Note that $X \sim N(40, 6^2)$. Consequently, we have

$$q^* \ = \ \mu \ + \ z^*\sigma \ \Rightarrow \ 49 \ = \ 40 \ + \ (z^*)(6) \ \Rightarrow \ z^* \ = \ 1.5$$

Then, for the critical ratio we have

$$z^* \ = \ 1.5 \ \Rightarrow \ \Phi \ - (1.5) \ = \ 0.9332 \ \Rightarrow \ F(q^*) \ = \ 0.9332$$

Note that the unit overage cost is given by $c_o \ = \ \$0.35$/unit. Hence, for the implied unit underage cost, we have

$$F(q^*) = 0.9332 \Rightarrow \frac{c_u}{c_u + 0.35} = 0.9332 \Rightarrow c_o = \$4.89/\text{unit}$$

5. The demand for pressure washers at Clyde's U-Rent-It is shown in the following table.

Demand	Frequency
0	0.30
1	0.20
2	0.20
3	0.15
4	0.10
5	0.05

The washers are rented by the day only. The profit on the pressure washers is $10 per day. Currently, Clyde has 4 pressure washers.

a. Assuming that Clyde's ordering decision is optimal, what is the implied range of unit overage cost per washer?
b. Your answer from part (a) has been presented to Clyde, who protests that the amount is too low. Does this suggest an increase or a decrease in the number of pressure washers he stocks? Explain.
c. Suppose now that the $10 mentioned as profit is instead the unit overage cost per day per washer and the unit underage cost is unknown. Assuming that the optimal number of washers is four, what is the implied range for the unit underage cost per washer?

Solution:

a) In this case, it seems that the unit cost of underage is equal to the profit, i.e., $c_u = \$10/unit$. The unit cost of overage is not known.
If $q^* = 4\ units$, then we have $0.85 \leq F(q^*) < 0.95$ based on the cumulative p.d.f.

Consequently, we have

$$0.85 \leq \frac{10}{10 + c_o}$$
$$8.5 + 0.85c_o \leq 10$$
$$c_o \leq \$1.76/\text{ unit}$$

and

$$\frac{10}{10 + c_o} < 0.95$$
$$10 < 9.5 + 0.95c_o$$
$$c_o > \$0.53/\text{ unit.}$$

b) If c_o is greater than the range found in part (a), then it suggests that the number of pressure washer stocked must decrease. We have $F(q^*) = \frac{c_u}{c_u + c_o}$. Everything else being equal, i.e., $c_u = \$10/\text{unit}$, an increase in c_o leads to a decrease in $F(q^*)$, i.e., the optimal order quantity decreases.

c) In this case, it seems that the unit cost of overage is equal to the profit, i.e., $c_o = \$10/\text{unit}$. The unit cost of underage is not known.
If $q^* = 4$ units, then we have $0.85 \leq F(q^*) < 0.95$ based on the cumulative p.d.f. Consequently, we have

$$0.85 \leq \frac{c_u}{c_u + 10} \Rightarrow 0.85c_u + 8.5 \leq c_u \Rightarrow c_u > \$56.67/\text{unit}$$

and

$$\frac{c_u}{c_u + 10} < 0.95 \Rightarrow c_u < 0.95c_u + 9.5 \Rightarrow c_u < \$190.00/\text{unit}$$

6. Rockwell Collins (RC) is a manufacturer of control boards for plane cockpit systems. RC has recently decided to discontinue the manufacturing of a particular control board that is currently being used by a small fleet of jet planes. RC must decide how much to make with the last manufacturing run before this board is discontinued. RC currently has zero units of this board in stock. Based on the past sales data, the demand over the lifetime of the product for the existing fleet of planes can assumed to be a Poisson distributed random variable with a mean of 3 units. The unit underage and overage costs are estimated to be $1,000 and $100 per unit, respectively. The cu parameter is large because a stock-out will disappoint customers and because the product will not be manufactured again. What is the optimal lifetime buy quantity for this control board?

Solution: In this case, the unit overage cost is estimated to be $c_o = \$100/\text{unit}$. Similarly, the unit underage cost is estimated to be $c_u = \$1,000/\text{unit}$. Then, the critical ratio is given by

$$F(q^*) = \frac{1000}{1000 + 100} = 0.91.$$

We have $F(q^*) = P\{X \leq q^*\} = 0.91$. Note that $X \sim Poisson(3)$. In order to find the optimal value of q^* in this case, we plug in different values of q in the c.d.f. to find the value of q such that $P\{X \leq q^*\} \geq 0.91$, since in this case the demand is a discrete random variable. Consequently, we have

$$F(0) = \frac{e^{-3}3\text{^}0}{0!} = 0.049$$

$$F(1) = F(0) + \frac{e^{-3}3\text{^}1}{1!} = 0.199$$

$$F(2) = F(1) + \frac{e^{-3}3\text{^}2}{2!} = 0.423$$

$$F(3) = F(2) + \frac{e^{-3}3\text{^}3}{3!} = 0.647$$

$$F(4) = F(3) + \frac{e^{-3}3\text{^}4}{4!} = 0.815$$

$$F(5) = F(4) + \frac{e^{-3}3\text{^}5}{5!} = 0.916$$

Hence, we the optimal order quantity is $q^* = 5$ boards.

7. A retailing firm is to purchase swimsuits for the summer season. The firm buys its swimsuits from a low-cost provider in Asia but is only able to make a single purchase per year. The selling price for the swimsuits is p = $20:00/unit. The firm pays $5.50 per unit. The firm can sell excess inventory outside North America for a salvage value of 2.75 per unit. The management believes that no significant goodwill is lost with a lost sale. For these swimsuits, little or no historical information about demand is available. Based on her experience, the purchasing manager believes that the demand distribution is not symmetrical. Furthermore, she believes that it would be appropriate to use a triangular distribution to approximate the demand distribution. She estimates the three parameters of a triangular distribution as follows: Minimum demand: $D_{min} = 3{,}000$ units, Most likely demand: $D_{mode} = 5{,}000$ units Maximum demand: $D_{min} = 9{,}000$ units.

a. What is the unit cost of overage?
b. What is the unit cost of underage?
c. What is the critical ratio?
d. Provide a sketch the estimated demand distribution.
e. What is the optimal order quantity for the swimsuits?

Solution:

a) Unit cost of overage is given by

$$c_o = 5.50 - 2.75 = \$2.75/unit.$$

b) Unit cost of underage is given by

$$c_u = 20 - 5.50 = \$14.50/unit.$$

c) Critical ratio is given by

$$\frac{14.50}{14.50 + 2.75} = 0.84.$$

d) A sketch of the pdf is provided below

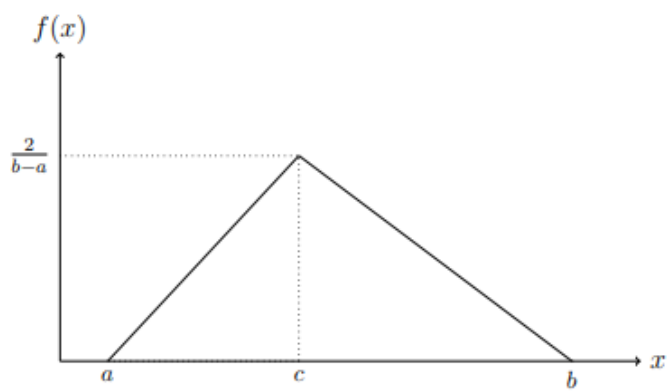

Figure 1: The plot of the p.d.f. for a triangular distribution with $a = 3,000$, $b = 9,000$, and $c = 5,000$.

e) Note that the p.d.f. of a triangular distribution is given by

$$f(x) = \begin{cases} 0, x < a \\ \dfrac{2(x-a)}{(b-a)(c-a)}, a \le x < c \\ \dfrac{2}{b-a}, x = c \\ 1 - \dfrac{2(b-x)}{(b-a)(b-c)}, c < x \le b \\ 1, b < x \end{cases}$$

and the c.d.f. of a triangular distribution is given by

$$F(x) = \begin{cases} 0, x \le a \\ \dfrac{(x-a)^2}{(b-a)(c-a)}, a < x \le c \\ 1 - \dfrac{(b-x)^2}{(b-a)(b-c)}, c < x < b \\ 1, b \le x \end{cases}$$

Consequently, we have

$$F(q^*) = 1 - \frac{(9,000 - q^*)^2}{(9,000 - 3,000)(9,000 - 5,000)} = 0.84 \Rightarrow q^* = 7,040.41 \text{ units}$$

8. Happy Henry's car dealer sells an imported car called the EX123. Once every three months, a shipment of the cars is made to Happy Henry's. Emergency shipments can be made between these three-month intervals to resupply the cars when inventory falls short of demand. The emergency shipments require two weeks, and buyers are willing to wait this long for the cars, but will generally go elsewhere before the next three-month shipment is due. From experience, it appears that the demand for the EX123 over a three-month interval is normally distributed with a mean of 60 and a variance of 36. The cost of holding an EX123 for one year is $500. Emergency shipments cost $250 per car over and above normal shipping costs.

a. How many cars should Happy Henry's be purchasing every three months?
b. Repeat the calculations, assuming that excess demands are backordered from one three-month period to the next. Assume a loss-of-goodwill cost of $100 for customers having to wait until the next three-month period and a cost of $50 per customer for bookkeeping expenses.
c. Repeat the calculations, assuming that when Happy Henry's is out of stock of EX123s, the customer will purchase the car elsewhere. In this case, assume that the cars cost Henry an average of $10,000 and sell for an average of $13,500. Ignore loss-goodwill costs for this calculation.

Solution:

a) The item in consideration is not perishable, and excess demand can be satisfied by emergency shipments. In this setting, the unit cost of underage is given by $c_u = \$250/\text{unit}$, and the unit cost of overage is given by $c_o = \frac{500}{4} = \$125/\text{unit}$. Note that in calculating the overage cost per quarter, we ignore the cumulative effect of the interest rate and simply

obtain the holding cost per quarter by diving the holding cost per year by four. Consequently, the critical ratio is given by

$$F(q^*) = 250\,250 + 125 = 0.6667$$

We have $F(q^*) = P\{X \leq q^*\} = 0.6667$. Note that $X \sim N(60, 6^2)$. Consequently, we have

$$F(q^*) = 0.6667$$

That is, we have

$$\Phi(z) = 0.6667 \Rightarrow z^* = 0.4308$$

Finally, we have

$$\begin{aligned} q^* &= \mu + z^*\sigma \\ &= 60 + (0.4308)(6) \\ &= 62.59\ cars \end{aligned}$$

b) Now, backordering for excess demand is possible, so Happy Henry's does not need to make emergency shipments. In this setting, the unit cost of underage is given by $c_u = 100 + 50 = \$150/unit$, and the unit cost of overage is given by $c_o = \frac{500}{4} = \$125/unit$. Consequently, the critical ratio is given by

$$F(q^*) = \frac{150}{150 + 125} = 0.5455.$$

We have $F(q^*) = P\{X \leq q^*\} = 0.5455$. Note that $X \sim N(60, 6^2)$. Consequently, we have

$$F(q^*) = 0.5455$$

That is, we have

$$\Phi(z) = 0.5455 \Rightarrow z^* = 0.1143$$

Then we have

$$\begin{aligned} q^* &= \mu + z^*\sigma \\ &= 60 + (0.1143)(6) \\ &= 60.69\ \text{cars} \end{aligned}$$

c) In this case, if Happy Henry's does not have the specific car upon customer arrival, the customer leaves, and the sales is lost. In this setting, the unit cost of underage is given by $c_u = 13,500 - 10,000 = \$3,500/\text{unit}$, and the unit cost of overage is given by $c_o = \frac{500}{4} = \$125/\text{unit}$. Consequently, the critical ratio is given by

$$F(q^*) = \frac{3{,}500}{3{,}500 + 125} = 0.9655$$

We have $F(q^*) = P\{X \leq q^*\} = 0.9655$. Note that $X \sim N(60, 6^2)$. Consequently, we have

$$F(q^*) = 0.9655$$

That is, we have

$$\Phi(z) = 0.9655 \Rightarrow z^* = 1.8184$$

Then we have

$$\begin{aligned} q^* &= \mu + z^*\sigma \\ &= 60 + (1.8184)(6) \\ &= 70.91\ \text{cars} \end{aligned}$$